

Translating English to symbolic logic:-

eg:- Recall Goldbach's conjecture

{ Every even number greater than two, is
the sum of two primes. }

$X \rightarrow$ even numbers greater than 2

$P \rightarrow$ all prime numbers

① $\forall n \in X, \exists p, q \in P, n = p + q$

② $(n \in X) \Rightarrow (\exists p, q \in P, n = p + q)$

eg:- at least one of the integers x and y is even.

① $E(x) \vee E(y)$, where $E(x)$ means x is even

② $(x \text{ is even}) \vee (y \text{ is even})$

$(x \text{ even}) \vee (y \text{ even})$

eg:- The integer x is even, but the integer y is odd.

① $E(x) \wedge \sim E(y)$, where $E(x)$ means

② $E(x) \wedge O(y)$, where $\sim$ means